MOHAMED HEDI BOUHOULA

Embedded Systems & Firmware Engineer

Ottawa, ON | bouhoulamohamedhedi11@gmail.com | Canadian Permanent Resident
linkedin.com/in/mohamed-hedi-bouhoula-a3877a114 | github.com/mohamedhedibouhoula

PROFILE

Embedded systems engineer with a Master's in Electrical & Automation Engineering (WES-assessed Canadian equivalent) and R&D experience building embedded prototypes. Built the robotic-arm and computer-vision control system for a medical device deployed with Hamad Medical Corporation, developed with Carnegie Mellon University Qatar. Now focused on **STM32 firmware in C** with the HAL, writing peripheral drivers (GPIO, interrupts, I2C) and publishing them on GitHub.

TECHNICAL SKILLS

- **Languages:** C, C++, Python
- **Microcontrollers:** STM32 (C0, F1, L0), ESP32-S3, Arduino, NVIDIA Jetson Nano
- **Firmware:** STM32 HAL, STM32CubeIDE / CubeMX, ESP-IDF, GPIO, EXTI interrupts, I2C, UART, register bit manipulation, driver design (.h/.c, encapsulation)
- **Tools:** Git / GitHub, Wokwi simulator, KiCad, Proteus, Linux
- **Other:** OpenCV, YOLOv8, sensor integration, motor control, robotics

PROJECTS

STM32 HAL I2C LCD Driver — C · STM32 HAL · I2C

github.com/mohamedhedibouhoula/stm32-hal-i2c-lcd-driver

- Wrote a HAL-only driver for a 16x2 LCD with a PCF8574 I2C backpack on an STM32C031 (no Arduino libraries).
- Implemented the 4-bit HD44780 init sequence and EN-pulse timing from the datasheets; configured I2C1 with STM32CubeMX.
- Structured as a reusable module (lcd.h / lcd.c, static private helpers, I2C handle passed by pointer) and documented the debugging of a nibble-alignment bug.

EV Wall-Charger Firmware (in progress) — C · STM32 · ADC · PWM · EXTI

- Controller firmware with a latched E-stop on an external interrupt, status LCD, and planned ADC current/temperature sensing, PWM fan control and a charging state machine.

Wi-Fi Presence Sensing (in progress) — C · ESP-IDF · Python

- Captures Wi-Fi Channel State Information on an ESP32-S3 (ESP-IDF), with Wi-Fi credentials stored in NVS and a serial command interface.
- Python bridge detects motion from the CSI stream and pushes live updates over WebSocket to a 3D web view.

PROFESSIONAL EXPERIENCE

Electronics Engineer – R&D — 2020 – 2022

Synaptic Vanguard — Doha, Qatar

- Built embedded prototypes in C++ on Arduino-class microcontrollers, integrating sensors, actuators and motor control.
- Developed the robotic-arm and computer-vision control system for a COVID-19 medical device deployed with Hamad Medical Corporation, in partnership with Carnegie Mellon University Qatar.
- Ran computer-vision models (OpenCV, YOLO) on NVIDIA Jetson Nano for real-time detection.

Technical Sales Engineer — 2022 – 2024

Chip International — Doha, Qatar

- Advised engineering customers on electronic components and solutions, translating technical requirements into product selections.
- Worked with suppliers on availability, pricing and technical documentation for customer projects.

Store Manager — 2024 – Present

KFC Store #1416 — Kanata, Ottawa, ON

- Lead a restaurant team: staff scheduling, training, food-safety compliance and daily operations.
- Build small software tools to improve operations, such as a staff scheduling web application.

Earlier roles (Doha, Qatar) — 2015 – 2020

Call Centre Agent, Echo Gulf (Barwa Bank project), 2016–2020 · IT Technician, Technoserve, 2015–2016

EDUCATION

Master's in Electrical & Automation Engineering — 2014

Polytechnic Institute of Sousse, Tunisia — WES-assessed as a Canadian Master's equivalent

Languages: English · French · Arabic